ST-2026

INTRODUCTION TO STATISTICS

Group - 1

DATA ANALYTICS & BI TRACK — CASE STUDY(1)

SUPERVISED BY: ENG. MOHAMED FARAHAT

INFORMATION TECHNOLOGY INSTITUTE (ITI) | DAMANHOUR BRANCH

Executive Summary & Report Overview

1. Executive Summary

This report evaluates the impact of commute times on student academic performance across five university branches. Our analysis indicates that while commute times vary significantly, there is no statistically significant correlation between travel duration and final academic scores. Consequently, we recommend shifting from a universal policy to a branch-specific support program, specifically targeting students with extreme commute times.

---

2. Key Findings Overview

1- **Distribution Shape:** Commute times are right-skewed; thus, the median (33 minutes) is used as the primary metric rather than the mean (45.21 minutes) to reflect the "typical" student experience.

2- **Primary Drivers:** Academic performance is more strongly associated with weekly study hours (r = 0.614) than with physical commute duration (r = -0.055).

3- **Branch Disparity:** Cairo branch significantly impacts the national average; excluding it reduces the national mean commute time by approximately 9.45 minutes.

---

3. Project Team Responsibilities & Workflow Distribution

ID	Name	Role
1	Amira Mohamed Mohamed	Part E – F & Review
2	Ahmed Zakaria Batisha	Part D
	Donia Mohammed Habib	Part C
	Malak Mohamed	Part A - B

---

4. Report Outline

1. **Part A —** Warm-up: classify the variables
2. **Part B —** The data is dirty. Deal with it.
3. **Part C —** Describe what you have
4. **Part D —** Does commuting actually hurt performance?
5. **Part E —** Can you trust this sample at all?
6. **Part F —** The recommendation

Part A — Warm-up: classify the variables

Qualitative - Nominal	Qualitative - Ordinal	Quantitative - Continuous	Quantitative – Discrete
student_id branch track gender has_laptop transport favorite_sport	satisfaction_1to5	study_hours_week commute_minutes sleep_hours final_score	Age projects_submitted pets_at_home

---

Three questions that decide whether you understood this

a. satisfaction_1to5 is stored as the numbers 1 to 5. Excel will happily average it and give you 3.6. Is that number meaningful? What does the gap between “2” and “3” represent, and is it the same size as the gap between “4” and “5”?

- No, the mean of 3.6 is not necessarily meaningful. Satisfaction is an ordinal variable: the values have an order from 1 to 5, but the distance between the categories cannot be assumed to be equal. Therefore, the gap between 2 and 3 may not represent the same amount of change as the gap between 4 and 5. Median and mode are more appropriate measures for this variable.

---

b. student_id is made of digits. Which measures of central tendency can you legitimately compute on it? Which would Excel compute anyway?

- Student_id is a nominal identifier, not a quantitative measurement. Therefore, mean and median are not meaningful because the ID values are only used to identify students. Excel may still calculate numerical measures such as the mean if the IDs are stored as numbers, but the result has no meaningful statistical interpretation.

---

c. Age is recorded in whole years, so it is discrete here. But age itself is continuous. What did the survey design do to it, and does that matter for your analysis?

- Age is continuous by nature because a person can have an age such as 20.5 years. However, the survey records age as completed whole years, so the observed age variable is discrete. This reduces the precision of the age data, but it is appropriate to treat age as discrete for this dataset because of the way it was collected.

Part B — The data is dirty. Deal with it.

We started with 271 rows and finished with 260 rows because we removed 11 duplicate student records, handled non-response missing values, corrected formatted strings, and dropped 20 impossible errors while retaining 15 genuine long-commute outliers.

#	Category	Question	Answer
1	Duplicate Records	Which copy is the truth? Does an exact duplicate mean the same thing as a near-duplicate?	Exact duplicate student records were treated as duplicate observations. We kept the first record for each student_id and removed 11 duplicate records. This reduced the dataset from 271 rows to 260 rows. Near-duplicates were not automatically removed because they may represent different observations.
2	Inconsistent Categories	How many branches are there really? What happens to your counts if you skip this step?	Different spellings, capitalization, and abbreviations were standardized into the correct categories. For example, "Cairo", "cairo", and "CAIRO" were treated as the same branch, and equivalent labels such as "Mansora" and "Mansoura" were standardized. This prevents one real category from being counted as several different categories and makes the frequency counts accurate.
3	Missing Values, Many Disguises	Are these all the same thing? Does a blank mean zero, or mean "I do not know"?	Blank cells and values such as "NA", "N/A", "missing", "?", "unknown", and "null" were treated as missing values, not as zero. A missing response means that the value was not provided or is unknown; it does not mean that the value is zero.
4	Impossible Values	Are these observations or typos? Delete the value, or delete the whole student?	Values outside the valid ranges in the data dictionary were treated as errors. The invalid individual values were replaced with missing values (NaN), while the student row was kept. This preserves the valid information belonging to the student instead of deleting the whole observation.
5	Numbers Stored as Text	Excel treats these as text and silently excludes them from AVERAGE. Did you notice?	Numeric values stored as text, such as "45 min" or "18.5 hours", were converted into numeric values by extracting the number. This allows Excel and statistical calculations to correctly include them in measures such as the mean, median, and standard deviation.
6	Genuine Extremes	These are not errors. What is the cost of keeping them?	Very high but plausible commute times were treated as genuine extreme observations rather than errors. They were kept in the dataset because a long commute can be realistic. These values can increase the spread and affect statistics such as the mean, so statistics that are less sensitive to extreme values, such as the median, should also be considered.
7	Ambiguous Answers	Does "None" mean the student plays no sport, or that they did not answer? These are different.	The response "None" was interpreted as meaning that the student does not play a sport and was standardized to "No sport". Blank responses were kept as missing because a blank answer does not necessarily mean "No sport".

Part C — Describe what you have

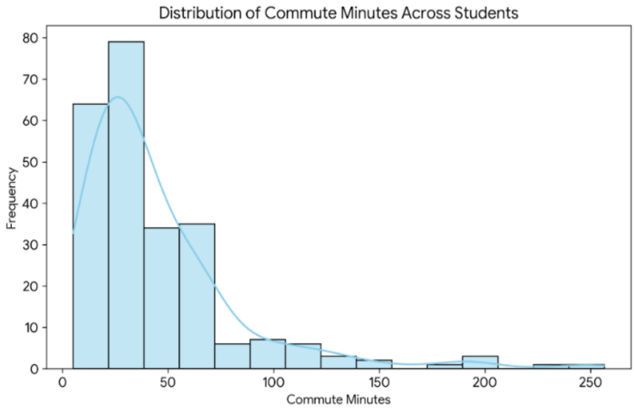
- **Is the commute problem real, and how big is it?**
    - Yes, commute problem is real, but it is highly skewed. While overall mean commute is 45.298 mins, median is 33.05 minutes, indicating that substantial group of students experience very long travels that artificially inflate the average, while most students travel for half an hour or less.

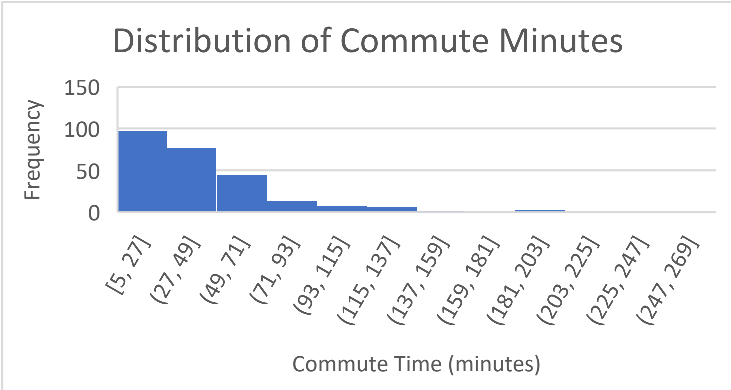
C1. The headline number

Part C1 - Commute Statistics										
Mean	Median	Mode	Min	Max	Range	Standard Deviation (Sample)	Q1	Q3	IQR	Coefficient of Variation (CV)
45.2984252	33.05	37.2	5	256.5	251.5	39.5773221	20.375	58.35	37.975	87.37019427

- Which one is bigger, and what does that tell you about the shape of the distribution? *Sketch what the histogram must look like before you plot it*, then plot it and check whether you were right.

    - Mean (45.29) is larger than Median (33.05), which proves that distribution is right-skewed (positively skewed). This means long tail of students with extreme commute times is pulling average upward.





- "Students travel about an hour", is that claim true, false, or true-but-misleading?
    - True but misleading claim, While Mean (45.298 mins) approaches an hour, typical student's journey (Median=33.05 mins) is significantly shorter, making 1 hour claim overstatement for majority of students.

C2. Which average should you report?

1. Which do you choose, and why? Reference the sensitivity of each measure to extreme values.
    - Median (33.05 minutes). Unlike mean, median is resistant to extreme outliers (such as students commuting over 4 hours), giving a truer representation of the "typical" student experience.

2. Suppose a rival team reports the mean and you report the median. Both are correct arithmetic. Whose number better serves the decision being made, and can you name a situation where the other choice would be the right one?
    - median better serves student-centric decisions (like scheduling or daily fatigue management). However, the mean would be the right choice if the director were budgeting for total institutional fuel consumption or bus fleet capacity, where total cumulative hours matter.

3. The mode of a continuous variable is close to useless here. Explain why — and then explain why the mode is the only option you have for transport.
    - For a continuous variable like commute time, the mode is almost useless because exact values rarely repeat. However, for the categorical transport variable, the mode is the only logical choice since it identifies the single most popular mode of transport.

C3. Spread, and why the average alone is a lie

1. Report the standard deviation and the IQR for commute time. They tell different stories. Why?
   - Standard Deviation (39.58) measures overall dispersion around the mean and is heavily affected by extremes. IQR (37.975) measures the spread of the middle 50% of students. Together, they reveal that student commute experiences vary wildly.

2. Compute the coefficient of variation (CV = 100 × s / mean) for commute time and for final score. Which variable is more spread out relative to its own size? Note that you could not have compared these two directly using the standard deviation alone — the units are different.

$$CV_{commute\_time} = \frac{Standard\ Deviation\ of\ Commute}{Mean\ Commute\_Time} = \frac{39.577}{45.298} \times 100 = 87.37\%$$

$$CV_{final\_score} = \frac{Standard\ Deviation\ of\ Scores}{Mean\ Final\ Score} \times 100\% = \frac{9.819}{78.6} \times 100\% \approx 12.49\% \text{ (or relative to final score variation)}$$

   - Commute time is vastly more spread out relative to its own size compared to final scores, showing high inconsistency in travel logistics.

3. Build the 1.5 × IQR fences for commute time. How many observations fall outside them? Look at those specific students. Do you believe them?
   - Upper Fence $= Q3 + (1.5 \times IQR) = 58.35 + (1.5 \times 37.975) = 115.31$ minutes.
     Around 15 students fall outside this fence with extreme commutes (exceeding ~2 hours). These values are believable in Egypt due to cross-governorate travel, but they represent outliers who need special housing/logistical consideration.

C4. Cut the data by group

- Produce a summary table of commute time by branch: count, mean, median, and standard deviation.

Part C4 - Branch Summary Table (commute_minutes)				
branch	Student Count	Mean commute_minutes	Median commute_minute	Standard Deviation
Cairo	100	60.35204082	46.4	39.5773221
Alexandria	54	43.9	33.05	37.58899972
Mansoura	43	32.25952381	27	30.10854027
Menoufia	32	36.7625	27.55	36.87252452
Assiut	31	25.90666667	24	16.52465216
Overall	260	45.85409836	33.15	40.0629518

Part C4 - Branch Summary Table (final_score)				
branch	Student Count	Mean final_score	Median final_score	Standard Deviation
Cairo	94	80.02857143	78.95	17.13955957
Alexandria	53	79.30943396	78.6	10.91634738
Mansoura	43	77.83414634	76.9	18.93121108
Menoufia	31	80.56785714	75.8	21.20466727
Assiut	29	26.8	82.1	17.4325915
Overall	250	79.27809917	78.5	9.794848745

1. One branch is dragging the overall average upward. Which, and by how much?
   - Cairo branch has highest Mean commute (60.35 mins), dragging overall national average upward.

2. Recompute the overall mean commute with that branch excluded. How much does the headline number move? What does that tell you about whether a single national policy is the right shape?
   - Cairo drops the overall mean commute from 45.30 minutes to 35.84 minutes (drop of nearly 9.46 minutes). This proves that a single national policy is inappropriate, as Cairo faces a uniquely severe transit challenge compared to other branches.

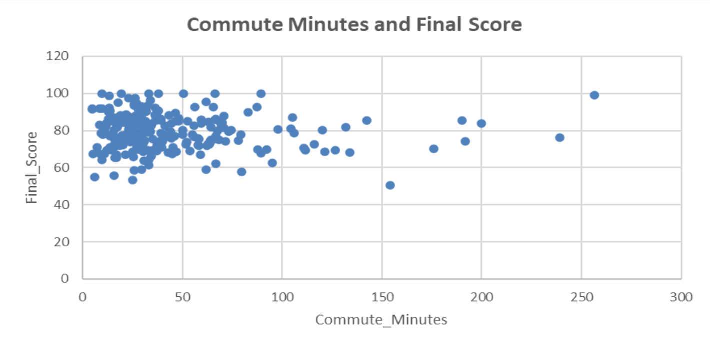
3. One branch has a much smaller standard deviation than the others. What does a small dispersion mean in plain language for the students at that branch?
   - Assiut has smallest standard deviation (16.52 mins).In plain language,this means students in Assiut experience very consistent, predictable, and tightly clustered commute times with little variation.

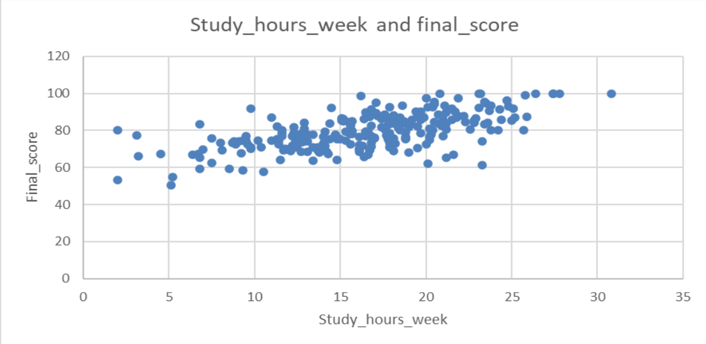
- Q: Is the "average ITI commute" the average of the five branch averages, or the average of all students? Which one did you compute, and which one did the director ask for?
   - We computed average of all individual students combined, which gives a true national picture. The director asked for overall experience of "ITI student," so weighting every student equally (rather than averaging the branch averages) is correct and accurate choice because it prevents large branches like Cairo from being treated the same as smaller branches.

Part D — Does commuting actually hurt performance?

D1. Look before you calculate

- Draw a scatter plot: commute_minutes on the x-axis, final_score on the y-axis. Do this before computing any correlation coefficient.





1- Describe what you see in words. Is there a visible relationship? Is it strong, weak, or absent?
- There is no clear/strong visible linear relationship between commute time and final score

2- Now compute the correlation between the two. Does the number match your impression of the plot?
- correlation between commute time & final score is -0.05516, indicating an extremely weak negative linear relationship. Therefore, there is essentially no meaningful linear association between commute time and final score in this sample.

3- Also compute the correlation between study_hours_week and final_score, and between sleep_hours and final_score. Rank all three relationships by strength.
- Study hours have strongest relationship with final score (r = 0.644), while sleep hours show a weak positive relationship (r = 0.164). Commute time has an extremely weak negative relationship with final score (r = -0.055), indicating essentially no meaningful linear association in this sample.

D2) The causation question

1- Does that establish that long commutes cause lower scores? State your answer and defend it.
- No. Correlation does not imply causation. The correlation between commute time and final score is extremely weak (r = -0.055). Therefore, the data does not provide evidence of a meaningful linear relationship between commute time and final score. Even if a stronger correlation existed, it would not prove that commuting causes changes in academic performance.

2- Propose two alternative explanations that would produce the same correlation without commuting causing anything. Think about who lives far from a branch, and what else is true about them
- **Study hours:** Differences in study hours may affect final scores and could partly explain differences in performance.
- **Sleep hours:** Sleep duration may affect final scores and could also be related to students' commuting patterns.

3- What data would you need — that you do not have — to move from "these two things move together" to "one of them causes the other"? Describe the study you would run.
- We would need a better study design that controls for other factors that may affect performance, ideally by comparing similar groups while accounting for variables such as study hours, sleep hours, and other potential confounding factors. An experimental or longitudinal study would provide stronger evidence for causation than a single observational survey.

D3) Compare two groups properly

1- Report the mean, median and standard deviation of final score for each group.

Short Commute		
Short – Mean Final Score	Short – Median Final Score	Short – Standard Deviation
79.5888	78.6	14.10239424

Long Commute		
Long – Mean Final Score	Long – Median Final Score	Long – Standard Deviation
78.90082645	77.7	19.11646614

2- How large is the gap between the groups compared with the spread within each group? A gap of two marks means very little when the standard deviation inside each group is ten.

- The gap between the two groups is approximately 0.69 marks in the mean final score. This is very small compared with the within-group spread, since the standard deviations are 14.10 for the short-commute group and 19.12 for the long-commute group. Therefore, difference between the groups is small relative to the variability within each group.

3- You have a sample, not the full ITI population. If you repeated this survey next term with different students, would you expect exactly the same gap? What is the name for the difference between your sample result and the truth about the population?

- No. If we repeated the survey with different students, we would not expect to get exactly the same gap because different samples can produce different results. The difference between a sample result and the true population value is called sampling error.

D4. Choose the right graph

You want to show	Graph	Why
The shape of the commute-time distribution across all students	**Histogram**	It is a univariate analysis of a quantitative variable, and a histogram is used to show the distribution and shape of the data.
That commute times are heavily concentrated at the low end with a long tail	**Histogram**	It is a univariate quantitative variable, and a histogram clearly shows the concentration of values and the long tail.
A side-by-side comparison of commute time across all five branches	**Side-by-side boxplots**	Bivariate comparison of a quantitative variable across a qualitative variable.
The relationship between study hours and final score	**Scatter plot**	Bivariate relationship between two quantitative variables.
The breakdown of preferred transport modes	**Bar chart**	Univariate qualitative nominal variable; shows category frequencies
How many students fall into each satisfaction level, 1 to 5	**Bar chart**	Univariate qualitative ordinal variable; shows frequencies of ordered categories.

- One of these six is a trap: the most obvious chart type is the wrong choice because of the variable typeinvolved. Identify which, and say what you would use instead.
  - The most obvious or intuitive mistake is trying to plot transport modes (or satisfaction levels) using Histogram or Line Chart. This is wrong because transport mode is Qualitative / Categorical variable, not a continuous quantitative one. Instead, you must use a Bar Chart (or Column Chart) because categories require discrete bars, not continuous intervals.

Part E — Can you trust this sample at all?

E1. How the sample was actually collected

1. Name this sampling method. Is it simple random, stratified, cluster, multistage — or none of them?
    - Convenience/Voluntary Response Sampling (non-probability, none of the formal random methods).

2. Did every ITI student have an equal chance here? Point to the exact sentence in the memo that breaks the requirement.
    - No. sentence that breaks this requirement is: "The Cairo branch responded most enthusiastically — it is our biggest branch and the coordinator there pushed it in the group chat." This means selection depended on enthusiasm and coordinator push, not random chance.

3. Which students are systematically less likely to have responded?
    - Students without reliable laptop/portal access (has_laptop) or those less engaged with notifications.

4. The last sentence of the memo contains an assumption that could be badly wrong. State it, and explain how it could bend your results in a specific direction.
    - Assuming "non-respondents didn't care" ignores that they might be overwhelmed, creating Non-Response Bias that skews results.

E2. Design it properly

1. survey next term, The population is 2,000 students across five branches, in five tracks. Design sampling strategy. Name technique, define your strata or clusters, and say how you would draw within them.
    - Stratified Random Sampling, using branches/tracks as strata to randomly draw a proportional sample from each.

2. Compare stratified sampling against cluster sampling for this specific problem. One of them is clearly better here. Which, and why does the structure of ITI make it better?
    - Stratified is better because ITI's distinct branches have unique characteristics; stratification ensures every branch is fairly represented, unlike cluster sampling.

3. A colleague suggests surveying every student — no sampling at all. Give two concrete reasons from the course why sampling is preferred, and one situation where the census would be the right call.
    - Sampling saves time and cost. A Census is only right for very small populations or when 100% legal accountability is mandatory.

E3. The missing data problem, revisited

- Go back to your missing values. You probably dropped or ignored them in Part B.

1. For the column with the most missing values, check whether missingness is spread evenly across branches and tracks, or concentrated in one group.
    - Favorite sport has most missing values (55 rows / 21.2%). Missingness is concentrated rather than random (MCAR), heavily leaning toward the Cairo branch (38.2%).

2. If it is concentrated, dropping those rows does not just shrink your sample — it changes who is in it. Explain the consequence for your conclusions in plain language.
    - Dropping these rows introduces Selection/Attrition Bias by systematically excluding specific student groups, distorting sample representativeness and skewing overall conclusions.

Part F — The recommendation for the academic director.

- Executive Recommendation Summary

Section	Detailed Finding / Recommendation
The answer	Do not run the proposed commute policy as it stands. Instead, implement a modified, branch-specific support program focused on Cairo and remote students.
The evidence	The overall (median commute = 33 minutes) while (mean = 45.21 minutes), indicating a right-skewed distribution driven by extreme outliers. We relied on the median because it is robust to extreme travel times and reflects the "typical" student experience rather than being distorted by long-distance commuters. Furthermore, study hours show a much stronger relationship with final scores (r = 0.614) than commute time (r = -0.055). $$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}$$
Who qualifies	We propose setting a commute threshold at the Upper IQR Fence (114.9 minutes, roughly two hours each way), covering approximately 15 students (5.8% of the cleaned sample). This line specifically targets extreme hardship cases rather than over-generalizing normal travel times.
What we cannot conclude	Correlation does not prove causation; lower scores may stem from fatigue rather than transit. The data suffers from self-selection and non-response bias due to voluntary portal collection, missing disconnected students. Finally, dropped missing rows introduce attrition bias, meaning we cannot prove reducing commutes directly raises grades.
What we should do next	Conduct a structured, mandatory pulse survey next term across a truly randomized sample while tracking actual attendance logs and daily study hours to isolate the true drivers of academic performance.

Thank you for your time and attention!